

THERMAL IMAGING For Industrial Testing

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This article explains the role of thermal imaging cameras in industrial applications, followed by their comparison with IR thermometers, working principle and buying tips

For manufacturing and other industries using heavy electrical and mechanical machines, thermal imaging is a one-time investment in continuous machinery testing for production level and efficiency. Thermal imagers are one of the most valuable diagnostic tools for industrial applications. These identify anomalies that are invisible to the human eye, thus allowing necessary corrective measures to avoid costly system failures.

Thermal imaging cameras record the intensity of infrared wave radiation (part of EM waves) and convert it into a visible image. This helps users to find out when and where maintenance is required. Electrical and mechanical installations tend to heat up before failure. By identifying those hot spots with a thermal imaging camera, preventive actions can be taken. This can avoid costly production breakdowns or, even worse, fire. Thermal imaging cameras can scan temperature distribution throughout the entire surface of machinery and electrical equipment without requiring contact.

Thermal imagers find application in almost every industry including defence, manufacturing, electronics, electrical, mobile, automotive and petrochemical. Applications for electrical systems can be divided into two categories: high-voltage installations and low-voltage installations. Thermal imaging cameras can be used for inspection of high-voltage substations, switchgear, transformers and outdoor circuit breakers from a safe distance without entering the risk zone. In low-voltage installations, thermal imaging cameras help to locate hot spots of loose connections, load imbalances, corrosion, increase in impedance to current, etc.

The temperature of mechanical components rises as these wear out and become less efficient. This increases overall temperature of equipment, causing these to fail. Thermal imaging cameras can monitor many mechanical systems including motors, couplings, gearboxes, bearings, pumps, compressors, belts, blowers and conveyor systems.

There are many other areas where thermal imaging cameras play an important role. These include flare detection, tank-level detection, hot spots detection in welding robots, aeronautical materials inspection, moulds inspection, and inspection in paper mills, pipe work in refractories and petrochemical installations.



Fig. 1: High-power transmission line inspection by thermal imaging cameras (Courtesy: www.test4less.co.uk)



Fig. 2: Modern thermal imaging cameras are small, lightweight and easy to use (Courtesy: www.50northspatial.org)

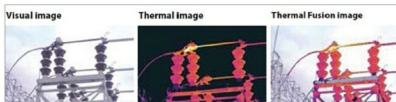


Fig. 3: Inspection of a substation revealing overheated components (Courtesy: www.bsria.co.uk)